Toxic Comment Detection

A Multi-Model NLP Pipeline for Multi-Label Classification of Toxic Online Comments

Institution:	University of Management and Technology (UMT), Sialkot Campus	**Course:**	Natural Language Processing (NLP)
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1. Executive Summary & Problem Definition

With over 3.8 billion social media users exposed to online discourse daily, automated moderation systems are crucial to scale. Manual moderation cannot keep up with the volume of text generated every minute. This project establishes a robust multi-label text classification pipeline that automatically flags comments across 6 distinct categories: `toxic`, `severe_toxic`, `obscene`, `threat`, `insult`, and `identity_hate`.

Unlike simple binary sentiment analysis, a single comment can belong to multiple categories simultaneously (e.g., being simultaneously obscene, insulting, and threatening). This project evaluates three architectures spanning traditional machine learning, sequential deep learning, and transformer architectures.

2. Dataset Snapshot & Class Imbalance Mitigation

The pipeline is trained and evaluated using the standard **Jigsaw Toxic Comment Classification Dataset** from Wikipedia talk-page comments:

- **Training Samples:** 159,571 comments
- **Testing Samples:** 153,164 comments
- **Distribution:** 78.0% of the dataset consists of completely clean (non-toxic) comments.

Severe Class Imbalance: Extreme imbalance exists in categories such as `threat` (0.30% positive rate) and `identity_hate` (under 1%). To prevent naive classifiers from optimizing purely for accuracy by predicting "clean" for all comments, specific algorithmic mitigations were implemented:

- **TF-IDF + Logistic Regression Baseline:** Evaluated with `class_weight='balanced'` to penalize minority class classification errors proportionally more.
- **BiLSTM Network:** Implemented a **custom weighted binary cross-entropy loss function** where each label's positive class is up-weighted by its negative-to-positive ratio (capped at 10x for stability).
- **Fine-tuned BERT:** Evaluated using dynamic threshold tuning to optimize for rare classes.

3. Pipeline Architecture

Model I: TF-IDF + Logistic Regression

Serves as a classical machine learning baseline. Text inputs are preprocessed (lowercasing, stopword removal, punctuation stripping, and tokenization) and vectorized via Term Frequency-Inverse Document Frequency (TF-IDF). A separate Logistic Regression model is fitted per label utilizing a One-Vs-Rest approach with balanced weights.

Model II: BiLSTM + Pretrained GloVe Embeddings

A sequential deep learning model. Word vectors are initialized using pretrained **GloVe 100-dimensional embeddings**. The architecture feeds these into a Bidirectional LSTM layer (64 units, dropout=0.3, recurrent_dropout=0.2) to capture forward and backward sequence dependencies, concluding with a Dense layer utilizing a Sigmoid activation for each of the 6 targets.

Model III: Fine-Tuned BERT Transformer

Leverages deep contextual representations from a pretrained `bert_en_uncased` model. Raw, uncleaned text is fed directly into the Hugging Face/TensorFlow BERT preprocessing layer (preserving structural and contextual casing signal), fine-tuned with an AdamW optimizer at a learning rate of 3×10^{-5}.

4. Comparative Performance Metrics

All three models were evaluated on the exact same held-out test set. While standard accuracy is reported, **Macro F1-Score** serves as the primary metric because it weights every label equally regardless of scarcity.

Model Architecture	Test Accuracy	Macro F1-Score	Core Advantage
TF-IDF + Logistic Regression	79.83%	0.4171	High speed, lightweight baseline, transparent features
BiLSTM + GloVe Embeddings	80.93%	0.4655	Captures sequential context, handles word order better
Fine-Tuned BERT Transformer	**86.79%**	**0.5875**	State-of-the-art bidirectional contextual understanding

5. Granular Error Analysis (Fine-Tuned BERT)

Below is the detailed per-class classification report and validation matrix obtained from the fine-tuned BERT model at its optimized inference threshold ($t = 0.35$):

[TOXIC] — Accuracy: 90.52%

True Positives: 5,443 | True Negatives: 52,467 | False Positives: 5,421 | False Negatives: 647

[SEVERE_TOXIC] — Accuracy: 98.91%

True Positives: 232 | True Negatives: 63,048 | False Positives: 563 | False Negatives: 135

[OBSCENE] — Accuracy: 95.37%

True Positives: 2,984 | True Negatives: 58,035 | False Positives: 2,252 | False Negatives: 707

[THREAT] — Accuracy: 99.56%

True Positives: 137 | True Negatives: 63,560 | False Positives: 207 | False Negatives: 74

[INSULT] — Accuracy: 96.41%

True Positives: 2,433 | True Negatives: 59,251 | False Positives: 1,300 | False Negatives: 994

[IDENTITY_HATE] — Accuracy: 98.90%

True Positives: 479 | True Negatives: 62,795 | False Positives: 471 | False Negatives: 233

> **Inference Engine & Live Demo:** The project wraps the fine-tuned BERT model into an interactive notebook **Gradio Toxicity Checker** interface, providing live web application testing. It renders text probabilities color-coded instantly with a visual toxicity verdict banner.

6. Conclusion

The experimental results demonstrate that while traditional methods (TF-IDF + LR) provide rapid execution speeds, they struggle to capture nuanced contextual signals. Incorporating sequential context via BiLSTM improves the Macro F1-score. Ultimately, the **Fine-Tuned BERT Transformer** yields the strongest performance, establishing a robust tool for commercial-grade automated content moderation.